DAY 15 — Data Visualization for Forecast Results & Model Interpretation

Day 15 focused on interpreting the forecasts created using Prophet. A model is only useful if we can interpret and communicate its results effectively. Today's session improved my skills in data storytelling and result interpretation.

🔹 1. Plotting Model Forecasts

Prophet provides built-in visualizations:

model.plot()

model.plot_components()

These charts helped visualize:

- The predicted line
- Uncertainty intervals
- Weekly seasonality patterns

🔸 2. Comparing Actual vs Predicted Values

We plotted:

- Actual closing prices
- Forecast values
- Confidence bands

This comparison helped evaluate:

- Accuracy
- Error margins
- Trend correctness

🔸 3. Understanding Uncertainty Intervals

The model generated:

- Lower bound (yhat_lower)
- Upper bound (yhat_upper)

This provided realistic insight into:

- Minimum possible price
- Maximum possible price

- Risk assessment

🔸 4. Trend Analysis

The trend component revealed:

- Steady upward momentum
- Change points where growth accelerated
- Periods of slower trend movement

Tesla stock showed strong growth after mid-2020.

🔸 5. Weekly Seasonality Interpretation

The model identified:

- Monday & Friday as high-volatility days
- Mid-week stability
- Weekend closure effects

This taught me how trading behavior affects stock prices.

🔸 6. Final Forecast Visualization

Using Plotly, we created interactive charts showing:

- Hover details
- Zooming
- Real-time comparison

These visualizations were highly professional and suitable for presentations.

Key Takeaways from Day 15

- Visualization strengthens interpretation
- Uncertainty intervals are essential for financial forecasting
- Prophet's components make understanding easy
- Trend and seasonality provide actionable insights